
Vraic to the Future

VRAIC FUTURES ENGINE

Conversational, probabilistic population scenarios for Jersey



Ageing strains public finances and the workforce

JERSEY

≈ +700

net migration a year to keep the 16–64 population stable to 2080, at mid-range fertility¹

2029

Long-Term Care Fund exhausted on current plans²

2072

Social Security Fund exhausted under Budget 2026 (was 2082); about 10 years sooner with lower migration³

INTERNATIONAL

8% to 13%

UK health spending, % of GDP, 2030–31 to 2075–76, OBR baseline⁴

–12%

EU labour force, 2022 to 2070: 25 million people⁵

3.0 to 1.9

people aged 20–64 per person aged 65+ across the OECD, 2025 to 2050⁶

1. Statistics Jersey (2026). Jersey population projections 2025 to 2080, §2 and §5.2

2. Fiscal Policy Panel (2025). Annual Report 2025, p. 5

3. Government Actuary's Department (2026). Jersey Social Security Fund budget proposals impact

4. Office for Budget Responsibility (2026). Fiscal risks and sustainability, July 2026, p. 5

5. European Commission (2024). 2024 Ageing Report, Institutional Paper 279, p. 4

6. OECD (2025). Pensions at a Glance 2025, chapter 1, p. 14; reciprocal of 33 and 52 aged 65+ per 100 aged 20–64

Ageing reshapes every public service

WORKFORCE

2.1

people aged 16–64 per
person aged 65+ in 2050
3.2 in 2025

16–64	68,073 to 62,750
65+	21,188 to 30,380

HEALTH AND CARE

+26%

hospital bed days by 2040
62,560 to 78,620

65+ care claims, 2040	+43%
GP appointments, 2040	+7%

EDUCATION

–21%

primary pupils by 2035
7,113 to 5,594

Secondary, 2040	–26%
Aged 0–15, 2040	–24%

HOUSING

2,680

extra homes needed by
2040

90% range	210–5,190
Population, 2040	+1%

How it works

Ask

local LLM drafts the change

Review

validated assumptions, you confirm

Simulate

10,000 futures: births, deaths,
migration, rates

Answer

median, 90% range, chance

"Keep everything else the same, but make migration zero after 2030."

62,100

aged 16–64 in 2040
90% range 57,800–66,500

–3,500

against the +400 a year baseline

99%

chance of fewer people of working
age than at end-2025 (68,073)

Same cohort-component method as Statistics Jersey

Reproduces their projections

10,000 simulations per question

Ranges checked against what happened

Economy, health, schools, housing

ECONOMY AND SKILLS

"What is the chance of fewer than two people of working age per person aged 65+ by 2050?"

1 in 3 31% of 10,000 simulations; median 2.1 per person aged 65+; 90% range 1.9–2.3

HEALTH AND CARE

"How many nurses and midwives would 2040 need at today's staffing per bed day?"

1,397 FTE; 90% range 1,315–1,479; end-2025: 1,112

EDUCATION

"How many secondary pupils should we plan for in 2040?"

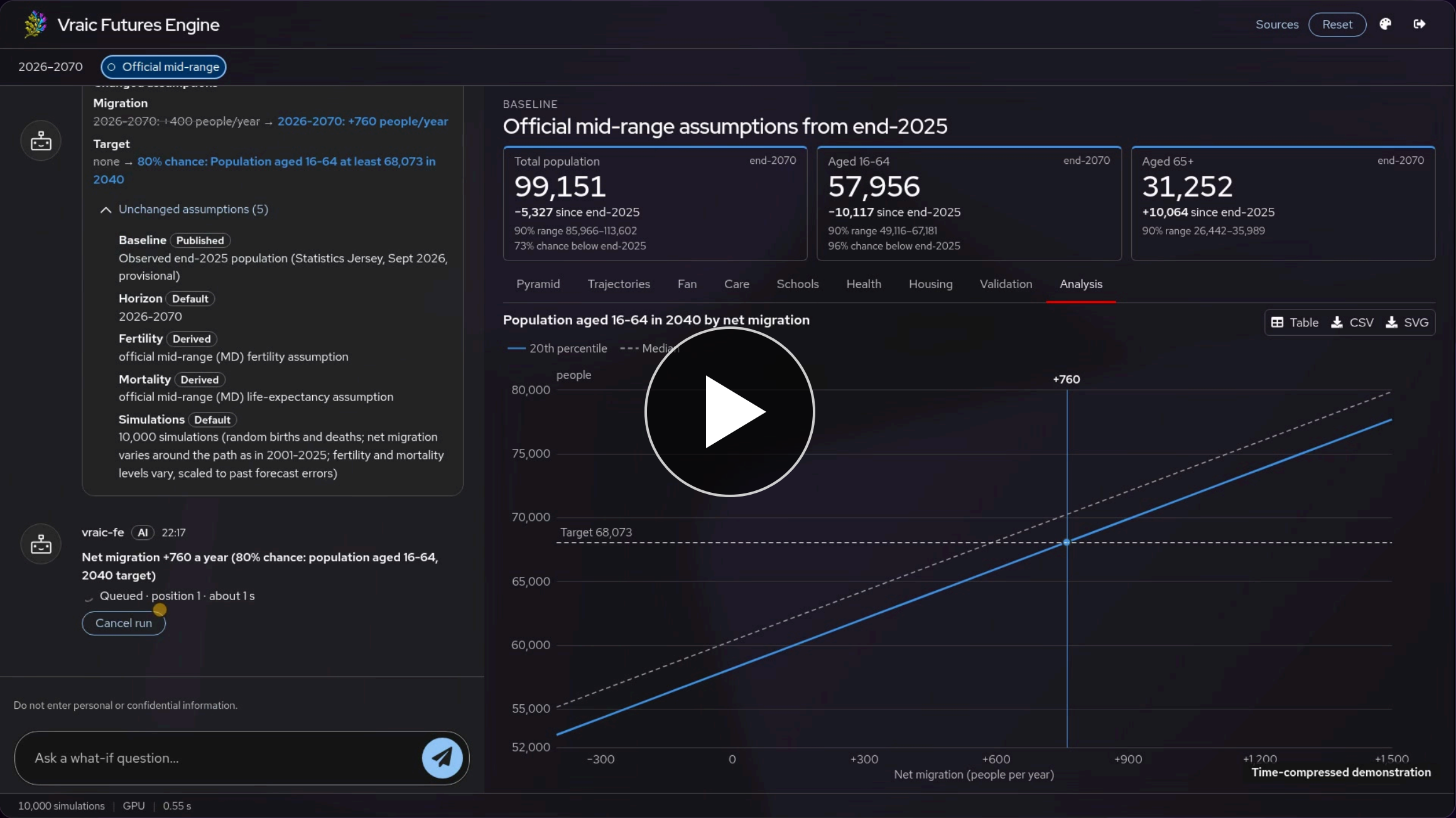
4,817 90% range 4,519–5,124; end-2025: 6,474

HOUSING

"What net migration is consistent with 2,000 additional homes by 2040?"

+314 a year

Demonstration



vraic.nat.je

ACCESS CODE

quartz-raven-982

A six-week pilot

WEEKS 1-2

Agree the baseline and methods with Statistics Jersey: statistical review

WEEKS 3-4

Trial with users from education, health, housing and other departments

WEEKS 5-6

Connect the Government's Copilot through an MCP server: every employee sees how demographics affect their department over time

ONGOING

Add datasets as users find what the modelling in their domain needs



VRAIC FUTURES ENGINE

github.com/gnathoi/vraic-fe